Code No: 118BA **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2021
DATABASE SECURITY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) Discuss various security threats to a database and explain the protection requirements for a database with suitable examples.
b) Define inference channel and discuss inference channels in a database system. [7+8]

2.a) Explain Take-grant model as an extension of access matrix model.
b) Illustrate the use of dynamic graph and static graph in Acten model. [8+7]

3.a) Compare the advantages of Biba model with Bell-LaPadula model.
b) What is meant by polyinstantiation? With examples describe polyinstantiated tuples and polyinstantiated elements. [7+8]

4.a) Contrast implicit and explicit flow to an object in lattice model.
b) Compare the security features of UNIX with OS/400 system. [8+7]

5.a) Which security model has been integrated into Xenix operating system? Explain the interpretation of this model in secure Xenix.
b) How are logical objects and dynamic objects are handled in database security? Give examples. [8+7]

6.a) Illustrate statistical database queries and describe the architecture of statistical security management facility for securing statistical database.
b) List the basic elements of intrusion detection system. [8+7]

7.a) What is a frame based system? Describe the security threats to this system.
b) Explain the concepts of semantic inheritance and security inheritance in object oriented database system. [8+7]

8. What is an active database? What are its special features? Does multilevel secure model suits active databases? Justify your answer with necessary scenarios. [15]

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